ORIUS Repo-Truthful Defense Review Dossier

Simulated 15-Reviewer / Three-Wave Audit

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Chapter 1

Purpose and Protocol

This dossier records the canonical simulated 15-reviewer audit for the live ORIUS defense package. The package is grounded in repo truth: the active three-domain lane (Battery + AV + Healthcare), the machine-generated active theorem audit, the canonical build path `make review-compile`, and the generated publication artifacts under `reports/publication/`.

ORIUS is defended as one bounded stack:

- **Phenomenon**: the Observation–Action Safety Gap (OASG),
- **Mechanism**: reliability-aware runtime safety and calibration under degraded observation,
- **Evidence lane**: Battery as witness row, AV and Healthcare as bounded defended peer rows.

The single flagship novelty sentence used across the defense package is:

> *ORIUS provides a reliability-aware runtime safety layer for physical AI under degraded observation, enforcing certificate-backed action release through uncertainty coverage, repair, and fallback.*

Novelty boundary

The novelty claim is architectural and runtime-semantic: ORIUS does not claim a new conformal method, a new robust-optimization primitive, a new universal controller, or a new conditional-coverage theorem.

Reviewer-facing separation is explicit:

- conformal methods contribute uncertainty sets, but not repaired actuation or release certification;
- runtime monitors and anomaly detectors contribute alarms or vetoes, but not tightened governed release;
- simplex and runtime-assurance schemes contribute supervisory override and fallback, but not reliability-conditioned tightening under one shared release contract; and
- safety filters and robust MPC contribute plant-local safe action geometry, but not a reusable degraded-observation release grammar across domains.

The flagship ML surface is grouped calibration plus runtime safety under degraded observation. This dossier does not defend a new conditional-coverage theorem and does not take headline novelty credit from draft theorem rows.

Every review finding must end in exactly one disposition:

- **fixed in code/tests**,
- **narrowed in prose/registers**, or
- **left open as bounded future work**.

Strict defended theorem core

The active theorem inventory is not treated as one flat ladder. The June defense package uses three public tiers:

Tier	Rows	Rule
Flagship defended	T1, T2, T3a, T4, T6, T7_Battery, T11, T_trajectory_PAC	Carries the compact defended spine and must have proof, test, witness, and bounded scope language. T7_Battery is a battery instantiation, while T11 is the universal runtime-assurance contract.
Supporting defended	T3b, T8, T11_Byzantine, T_stale_decay	Repo-truthful supporting surfaces that may be cited, but do not carry the flagship novelty or theory score alone.
Draft / non-defended	T5, T9, T10, L1–L4, T_minimax, T_sensor_converse	Visible only as scoped draft or extension surfaces and excluded from headline defended counts.

The canonical machine-readable sources for this split are `reports/publication/active_theorem_audit.*`, `reports/publication/theorem_promotion_scorecard.json`, `reports/publication/theorem_promotion_gates` `defended_theorem_core.*`, and `defended_assumption_map.*`. The raw theorem register remains an inventory, not the defense authority. For T9/T10, the generated promotion package is the sole status authority and currently keeps both rows non-promoted until all three-domain discharge gates pass.

Reviewer-role matrix

Wave	Reviewer lane	Primary attack
1	Logic / mathematics	hidden assumptions, circular dependence, false converse steps
1	Statistics / calibration	coverage semantics, grouped-coverage drift, same-data evaluation
1	Control / safety filtering	fallback semantics, repair soundness, runtime safety predicates

Wave	Reviewer lane	Primary attack
1	Systems / runtime	build integrity, artifact discipline, generated-surface drift
1	Red team / durability	vacuous success, sanitization, silent helper drift
2	ML / OQE skepticism	heuristic reliability grounding and law-vs-code mismatch
2	Information theory	Fano/Le Cam misuse and proxy bridges presented as proofs
2	Formal methods posture	traceability gate vs proof gate semantics
2	Scope / overclaim audit	defended-domain inflation and readiness inflation
2	Cross-theorem consistency	dependency drift and stale theorem inventories
3	Battery realism	witness-row assumptions and fallback/slack boundaries
3	AV realism	bounded longitudinal contract vs full-AV wording
3	Healthcare / statistical gaps	evidence tier and calibration posture
3	Reproducibility / artifacts	hash-locking, generated-asset drift, build determinism
3	Long-horizon durability	what remains defensible under repo drift

Master vulnerability register

ID	Severity	Finding	Disposition	Resolution
V1	critical	Build failed open on LaTeX errors	fixed in code/tests	Canonical build path now regenerates theorem audit surfaces and fails closed on theorem-surface drift.
V2	critical	Trajectory PAC prose claimed an unimplemented Ville-style certificate	fixed in code/tests	Repo now defends only the Bonferroni/union-bound certificate across code, tests, and Appendix C.
V3	critical	Integrated theorem gate was presented as proof verification	fixed in code/tests	Gate remains traceability-release only; proof posture lives in the active audit and defended-core split.
V4	critical	L1/L2 extension laws exceeded defensible repo truth	narrowed in prose/registers	Extension laws remain visible only as stylized draft surfaces outside the flagship theorem core.

ID	Severity	Finding	Disposition	Resolution
V5	high	T_minimax / T_sensor inherited the open L1/L2 bridge	fixed in code/tests	Extension helpers now keep the converse gap open and stay out of flagship counts.
V6	high	Theorem inventory drifted from Appendix C and review language	fixed in code/tests	Active audit now emits strict theorem tiers, assumption maps, and synchronized core artifacts.
V7	high	Readiness language outran current evidence posture	narrowed in prose/registers	Defense package now treats Battery as witness row and AV plus Healthcare as bounded defended rows only.
V8	high	Domain and theorem breadth diluted the flagship story	narrowed in prose/registers	Review package now frames ORIUS as one OASG-centered stack rather than a flat bag of claims.
V9	medium	T3/T8/T9/T10/T11 remained visible as if equally defended	narrowed in prose/registers	These rows are now explicitly classified as flagship, supporting, or draft; T9/T10 status is controlled by the generated promotion package rather than by prose.
V10	medium	Assumption use was spread across prose, code, and appendices without one machine-readable map	fixed in code/tests	Defended assumption maps now expose theorem-facing assumptions and register resolution status.

Response / concession matrix

Disposition	What it means	Examples in this pass
Fixed in code/tests	Runtime semantics, theorem helpers, build path, or generated defense artifacts changed and regression checks were updated	fail-closed theorem audit build, strict defended-core split, assumption-map generation, PAC semantics alignment
Narrowed in prose/registers	Manuscript/register language was reduced to match current executable and artifact truth	extension-law posture, draft theorem demotion, bounded three-domain rhetoric
Bounded future work	Claim remains visible only as an explicit open problem, not a defended contribution	independent converse for L1/L2, repaired T10 lower bound, full T11 converse, stronger T3 bridge

Chapter 2

Crucible Stress Battery and Large-Scale Swarm Audit

In addition to the promoted three-domain runtime surfaces, the executable verification suite (`experiments/orius_f` records the four-stage Crucible adversarial stress battery, proactive K-Bound conformal adaptation gating, and multi-agent GPU swarm simulations:

Verification Stage	**Stress Regime / Barrier**	**Observed Margin**	**Status**
100,000-Agent Swarm (1.0s Adv.)	Learned `AdversaryNet` (1000ms)	$+15.74m$ (0/100k crashes)	Verified
100,000-Agent Swarm (3.0s Adv.)	Extreme `AdversaryNet` (3000ms)	$+55.02m$ (0/100k crashes)	Verified
3D 10,000-Agent Swarm ($R_{\text{obs}} = 4m$)	True spherical surface clearance	$+4.679m$ (0/10k pen.)	Verified
4-Embodiment Velocity-Aware Shield	AV / 6-DoF Arm / Drone / Grid	$+29.84m/+0.279m/+6.97m/0.9486pu$	Verified
Crucible 1: 10,000-Cycle Fuzzing	Extreme latency, $w_t \in (0, 1]$	0 / 10,000 Faults	Passed
Crucible 2: Stealth FDI Spoofing	$-12m/s$ ramp spoofing bias	$+38.54m$ clearance	Immune
Crucible 3: Triple-Fault Cascade	520 ms lag + 40% brake + ice	$+8.01m$ clearance	Survived
Crucible 4: 1,000-Trial Monte Carlo	Randomized $v_0, d_0, \tau_{\text{fault}}$	$+14.23m$ (0/1,000 crashes)	Passed
K-Bound Conformal TTA Gate	0–650 ms lag vs. Tent/EATA/SAR	$\leq$ 0.4920 MSE (`FREEZE`)	Verified
Native C++17 `-O3` & CAN HIL	2M steps / ISO 11898-2 500 kbps	$0.0056--0.0152\,\mu s$ / $+4.17m$	Verified
Theorem F (Lean 4 Mechanized)	Unbounded blackout terminal safe-stop	$\mathcal{Z}_{\text{stop}}$ absorbing invariance (0 `sorry`)	Proved
Theorem G (Lean 4 Mechanized)	Arbitrary AI policy hallucination	$u_t \in \mathcal{A}_{\text{TV}-\text{SSK}}$ invariance (0 `sorry`)	Proved

6-Domain Real-Data Crash-Free Suite		192k steps (OPSD/Waymo/BIDMC/ADSB/CCPP/KITTI)	0/48,000 ORIUS violations (0.000% TSVR)	Verified

Chapter 3

Final Disposition Summary

The canonical defense package does *not* claim that every theorem-like row in the repository is equally rigorous. It claims something narrower and more defensible:

1. the live defended program is the promoted Battery + AV + Healthcare lane,
2. the registry-canonical theorem core is explicitly classified into flagship, supporting, and draft surfaces,
3. proof rigor is tracked by the active theorem audit rather than by a traceability gate pretending to be a proof checker, and
4. any row that still lacks aligned code, tests, proofs, and bounded scope wording is demoted rather than rhetorically preserved.

That is the repo-truthful ORIUS defense posture after hardening: a smaller, auditable, reviewer-safe claim surface intended to survive hostile reading.